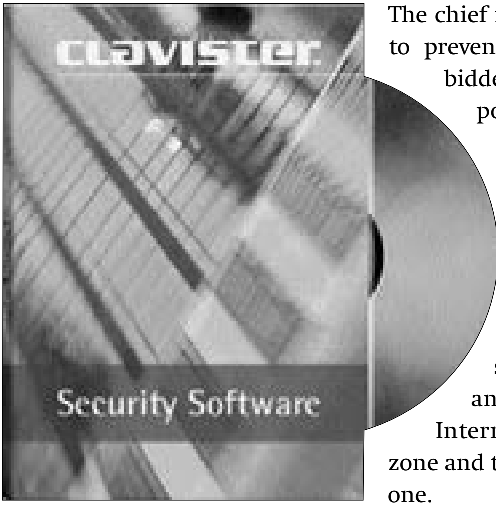
7.6 Firewalls



Firewalls have become a de-facto safety net for any network

The chief function of a firewall is to prevent communications forbidden by a certain security policy. Unlike an anti-virus, a firewall can be either hardware or software. The basic duty of a firewall is to control traffic between two different 'zones of trust' such as the Internet and an intranet, with the Internet being a no-trust zone and the intranet a high-trust one.

There are roughly three types of firewalls:

Personal Firewalls

These are software that filters through information entering or leaving a single computer. Network Firewall is a firewall setup on a dedicated network device or computer positioned on the boundary of two or more networks.

Network and Application layer firewalls

A network layer firewall works at a low level of the TCP/IP protocol stack. This means only those data packets that comply with the predefined rules, or the rules set up by the network administrator, will be allowed to pass. Today, network firewalls are built into most computer operating systems and network appliances. Modern firewalls filter traffic based on many information packet attributes such as the source IP, destination IP, and destination services such as WWW or FTP.